

Total No. of Questions : 4]

SEAT No. :

PE203

[6580]-563

[Total No. of Pages : 2

**B.E. (Mechanical) (Insem)
INDUSTRIAL ENGINEERING**

(2019 Pattern) (Semester - VII) (402044D) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Use of electronic calculator is allowed.
- 4) Assume suitable data, if necessary.

Q1) a) Summarize different productivity improvement methods? Elaborate 5 “S” system in brief with its benefits? **[7]**

b) Explain the term industrial engineering? What is its importance? Also explain PDCA chart in details. **[8]**

OR

Q2) a) Explain the term Productivity? Also explain any two Productivity Measurement Models **[8]**

b) ABC Company Produces 1600 kg of Spanners by consuming 2200 kg mix of raw material for a particular period. For the next period, the output is 3400 kg by consuming 4350 kg of the raw material and for the third period, the output is increased to 3600 kg by consuming 4450 kg of raw material. Comment on increase or decrease of productivity for Second & Third period. **[7]**

Q3) a) Explain different types of recording techniques used in method study? Also explain Two handed process chart with suitable example. **[7]**

b) Following are the recorded times of a machining operation. The corresponding rating and relaxation allowances are given below calculate Normal Time & standard time. **[8]**

P.T.O.

Element	Observed Time (Min)	Performance rating	Relaxation Allowance (%)
1	0.15	85	13
2	0.05	85	13
3	0.55	90	10
4	1.00	95	12
5	0.1	90	13

OR

Q4) a) Explain the term “Performance rating”? Illustrate different performance rating Methods? [7]

b) The work study engineer carried out the work sampling study. [8]

The following observations were made for a machine shop

The time duration of study = 120 hrs

Total number of observations = 7000

Number of working activities = 1200

Ratio between manual to a machine elements 2:1

Average rating factor 120%

Total no. of jobs produced during study = 800 Units

Rest and personal time for the job = 12%

